

Q 14)

a) there are 16 different bits, so size is 16

b) Method name, parameter list

c) Encapsulation means keeping data private and only allowing access through public methods.

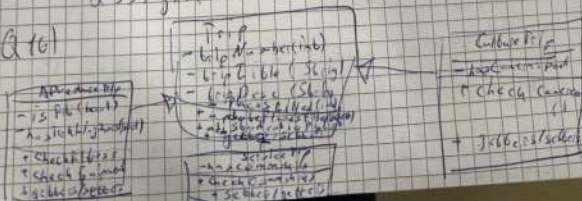
placesFilled, isRunning in Telp are private

d) After placesFilled is incremented, a check is made - if placesFilled has reached 10 or more, then isRunning is set to true

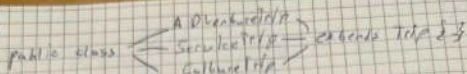
Q 15) a new field would need to be added to the Student object to store Student's ID field

b) Look up the friend's Student object verifying that the friend has been assigned to the same bit.

Q 16)



aii)



aiii) 1) Code reuse

2) Easier maintenance

aiii) it won't break other objects that rely on it

ii) Easier debugging and testing
Parallel development

Q (7)

c) ~~Trip 1, Trip 2, Trip 3, Trip 4, Trip 5, Trip 6, Trip 7, Trip 8, Trip 9, Trip 10~~

1) Set Trip 3 pointer to point to Trip 2

2) Set Trip 2's pointer to Trip 1

Q (8) aiii) FIFO → fairer

aiii) can't access a specific student's record

biii) For each trip that is a record

it adds all the student names from that trip's list into a list
ciii) more tedious